GAURAV JAIN

Surat, Gujarat, India | +91 7016503690 | gauravjain20001643@gmail.com
linkedin.com/in/gaurav-jain-18954a333 | github.com/LesssggooGauravv | leetcode.com/u/lesssggoogauravv

SUMMARY

ML Engineer who scaled multilingual TTS and voice cloning to 22 Indian languages in production at Jio Platforms and built an audio deepfake detector at 1.87% EER in a national hackathon. Deep experience in PyTorch, GANs, and edge-optimized deployment (TFLite), with full-stack shipping ability across Python, React, and Node.js.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, Bash
Tools & Core CS: Git, GitHub, Kaggle, Google Colab, Linux, Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, CNNs, GANs, TFLite, Model Quantization, RAG
Speech & Audio: ASR, TTS, Voice Cloning, Speaker Diarization, Mel Spectrograms, Noise Suppression, Librosa, Torchaudio, Gradio
Full-Stack: React.js, Node.js, Express.js, FastAPI, Tailwind CSS

EDUCATION

The LNM Institute of Information Technology (LNMIIT), Jaipur 2023 – 2027
B.Tech, Computer and Communication Engineering • CGPA: 7.3 / 10
Bachelor's Thesis: GAN-Based Deepfake Audio Detection (Ongoing), Supervised by Dr. Priyanka Gupta

EXPERIENCE

Machine Learning Intern – TTS for Indic Languages Jun 2026 – Jul 2026
Jio Platforms Limited, Navi Mumbai, India (On-Site)

- Engineered a 4-version Indic-to-English video dubbing pipeline (audio separation, enhancement, translation, TTS synthesis), improving output quality across 4 stakeholder-driven releases
- Architected a modular 6-stage system with automated per-segment QC gating and checkpoint-routing logic, enabling reliable selection across 5+ ML libraries and multiple language pairs
- Scaled cross-lingual voice cloning to 22 Indian languages using 100 reference speakers, generating a 6,250-sentence multilingual corpus and cutting inference time via multi-GPU round-robin dispatch
- Established a Hindi TTS QA workflow across 1,000+ rows with 130+ pronunciation mappings, streamlining manual review and catching domain/emotion mismatches across 73 categories
- Developed a multilingual speaker diarization framework (WhisperX, VoxLingua107, Pyannote), resolving Gujarati-Hindi script ambiguity and removing per-language dependencies for easier scaling
- Resolved dependency conflicts across 4+ production libraries and remediated hardcoded API credentials, improving system stability and security posture
- Evaluated 5 voice-cloning and emotion-transfer approaches for production readiness and documented TTS data formats for team-wide reuse

PROJECTS

VoiceShield – Offline Edge-Based Deepfake Voice Detection
Samsung Solve for Tomorrow Hackathon — Individual Submission

- Independently designed and built an offline, edge-deployable audio deepfake detector end-to-end for the Samsung Solve for Tomorrow Hackathon
- Trained a depthwise-separable CNN with SE-attention blocks on ASVspoof 2019 + WaveFake, achieving a 1.87% Equal Error Rate and 0.9982 ROC-AUC
- Compressed the network to a 76KB TFLite export for edge deployment, debugging a mixed-precision NaN-loss bug in distributed training
- Shipped a Gradio demo interface and reverted a fine-tuning change after catching a class-imbalance regression, protecting reliability

CNN-Based Acoustic Phoneme Classification Engine
Speech Processing / ASR Foundation

- Designed a 2D-CNN in PyTorch to classify self-recorded speech (22.05 kHz) into 7 phoneme categories, engineering a 3-channel time-frequency input from Mel-spectrogram + delta + delta-delta (128 Mel bands) features
- Built a leakage-safe train-test split protocol with controlled data augmentation to scale to ~100 samples/class, preventing speaker/recording overlap from inflating validation metrics
- Achieved 75–85% validation accuracy under strict non-leaking evaluation; confusion-matrix analysis confirmed errors clustered along genuine phonetic similarity, validating real acoustic structure was learned